

SIMPLICIAL COMPLEXES FORCING FAILURE OF THE WEAK LEFSCHETZ PROPERTY

1. BACKGROUND

You can find most of the necessary background in the [preliminaries.pdf](#) found on the [ISMaRT GitHub](#).

2. SIMPLICIAL HOMOLOGY

Recall that a simplicial complex can be used to model a topological space. These spaces have all sorts of interesting features, but some of the most prevalent ones are the “holes” that it has. It’s a bit tricky to mathematically define or quantify what a hole means, but it turns out that *homology* is a good way to do this. The intuition will be that the k -th homology group $H_k(X)$ whose *rank* (dimension) is the number of k -dimensional holes in X , where a k -dimensional hole is like a k -dimensional sphere.

Remark 2.1. This intuition measures up with what you might expect:

- $H_0 \rightsquigarrow$ connected components of X
- $H_1 \rightsquigarrow$ nontrivial circles or loops on X
- $H_2 \rightsquigarrow$ spheres inside X

Example 2.2. Consider the simplicial complex Δ which is a hollow triangle, i.e., the cycle graph C_3 . Everything is connected (one connected component), and there is a loop which you can’t “deform” to a point (it’s nontrivial in this sense). Moreover, the Δ isn’t really 3-dimensional, so you wouldn’t expect to fit any spheres inside of it. These observations are reflected through its homology groups:

$$\dim H_0 = 1 \qquad \dim H_1 = 1 \qquad \dim H_2 = 0.$$

Definition 2.3. The dimensions of the *reduced* homology groups $\tilde{H}_i$ are given by

$$\dim \tilde{H}_i = \begin{cases} \dim H_0 - 1 & \text{if } i = 0 \\ \dim H_i & \text{if } i \neq 0 \end{cases}.$$

(The utility here is that we can say, for instance, that (1) Δ is connected $\iff \dim \tilde{H}_0(\Delta) = 0$, and (2) Δ is *acyclic* $\iff \tilde{H}_i(\Delta) = 0$ for all $i \geq 0$; all for convenience).

There are many “theories” for determining these “holes”, but a decent one for computations in simplicial homology. This just means that our topological space can be modeled using a simplicial complex, and then we can do some linear algebra from the data of the simplicial complex to determine the homology groups. Here, the holes are measured via “boundaries”. To explicitly compute simplicial homology, we need the notion of the boundary map:

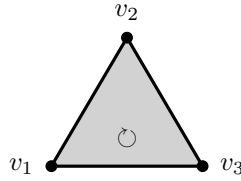
Definition 2.4. For a simplicial complex Δ , let C_k denote be the free module with basis given by the k -faces of Δ (technically, as the faces are sets, you must fix an orientation, but this is more for bookkeeping in computation). The elements of C_k are called k -chains. The boundary map $\partial_k: C_k \rightarrow C_{k-1}$ is a map taking k -faces to $(k-1)$ -faces, and is defined by

$$\partial(\{v_{i_0}, \dots, v_{i_n}\}) = \sum_{j=0}^n (-1)^j \{v_{i_0}, \dots, \hat{v}_{i_j}, \dots, v_{i_n}\},$$

where $\hat{v}_{i_j}$ means to exclude that entry.

Example 2.5.

Example 2.6. Consider the simplicial complex



oriented clockwise. Then

$$\partial(\{v_1, v_2, v_3\}) = \{v_2, v_3\} - \{v_1, v_3\} + \{v_1, v_2\}.$$

With our chosen orientation, this is saying that the boundary of the filled in triangle is comprised of following the triangle $v_2 \rightarrow v_3$, $v_1 \rightarrow v_3$ (but really in the reverse direction because of the minus sign, so $v_3 \rightarrow v_1$), and then $v_1 \rightarrow v_2$, thus tracing out the boundary like you would hope.

For the boundaries of the edges, you can do the boundary map calculation, but you should have the saying “head minus tail” in mind.

Much like we saw with free resolutions, it turns out that $\partial \circ \partial = 0$ (can you verify this?), so we get a chain complex

$$0 \rightarrow C_n \xrightarrow{\partial_n} \dots \xrightarrow{\partial_3} C_2 \xrightarrow{\partial_2} C_1 \xrightarrow{\partial_1} C_0 \rightarrow 0.$$

This means we can compute *homology* as $H_i(\Delta) = \ker \partial_i / \text{im } \partial_{i+1}$. We will only care about the *dimension* of the homology groups, so this means we need to compute $\dim \ker \partial_i$ and $\dim \text{im } \partial_{i+1}$.

We can represent these boundary maps with matrices as well since they are maps of free modules; you just have to fix a basis first. For instance, with the example above and only writing the indices:

$$0 \rightarrow C_2 \xrightarrow{\begin{matrix} & 123 \\ 12 & \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \\ 23 & \\ 13 & \end{matrix}} C_1 \xrightarrow{\begin{matrix} & 12 & 23 & 13 \\ 1 & \begin{bmatrix} -1 & 0 & -1 \end{bmatrix} \\ 2 & \begin{bmatrix} 1 & -1 & 0 \end{bmatrix} \\ 3 & \begin{bmatrix} 0 & 1 & 1 \end{bmatrix} \end{matrix}} C_0 \rightarrow 0.$$

This means that to compute $\dim H_i(\Delta)$, we can really just compute the ranks of these matrices with good old linear algebra and the rank-nullity theorem.

Exercise 2.7. Compute the boundary maps of the hollow octahedron from [preliminaries.pdf](#). Use a computer to compute the relevant ranks/nullities of the maps to then determine the dimensions of the homology groups.

3. COMPUTING BETTI NUMBERS OF STANLEY-REISNER RINGS

Let $R = \mathbb{K}[x_1, \dots, x_n]$ be a polynomial ring. We normally consider R with the *standard grading* where $\deg(x_i) = 1$ for all $1 \leq i \leq n$. However, we can also endow R with other gradings—usually called *multigradings*—which can carry finer information. For us, the *fine grading* on R is given by $\deg(x_i) = e_i \in \mathbb{Z}^n$ (e_i is the usual standard basis vector from linear algebra).

Example 3.1. Let $R = \mathbb{K}[x_1, x_2, x_3]$ with the fine grading. Then

$$\deg(x_1 x_2) = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \quad \text{and} \quad \deg(x_1^2 x_3) = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}.$$

Remark 3.2. We can always transfer from the multigraded setting to the standard graded setting. To do this, if $f \in R$ has multidegree $(a_1, \dots, a_n) \in \mathbb{Z}^n$, then $\deg(f) = \sum a_i$ in the standard grading.

The philosophy is that if something comes equipped with a multidegree, you can perform some nice combinatorics on it to get a cool result. From there, you can “specialize” (in some sense) to get the result you actually want back in the world of standard graded rings; the usual example here is graded Betti numbers.

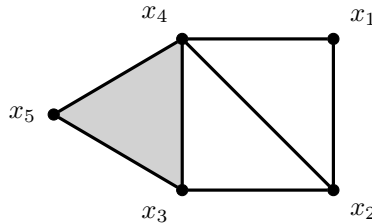
3.1. Hochster’s Formula. Let Δ be a simplicial complex, and let S/I_Δ be its Stanley-Reisner ring. Normally, to compute the graded Betti numbers of S/I_Δ , a graded minimal free resolution would have to be computed. A result of Hochster allows one to compute the graded Betti numbers of S/I_Δ purely in terms of the simplicial complex by computing either cohomology of subcomplexes of Δ or homology in the Alexander dual of Δ .

Definition 3.3. Let Δ be a simplicial complex on $[n] = \{1, \dots, n\}$. For a subset $W \subseteq [n]$, the *induced subcomplex* on W in Δ , denoted $\Delta|_W$ is the simplicial complex consisting of all faces of Δ whose vertices lie in W .

Theorem 3.4. Let Δ be a simplicial complex on $V = \{x_1, \dots, x_n\}$ with Stanley-Reisner ideal $I_\Delta \subseteq \mathbb{K}[x_1, \dots, x_n]$. Then for all i and j ,

$$\beta_{i,i+j}(S/I_\Delta) = \sum_{\substack{W \subseteq V \\ |W|=i+j}} \dim_{\mathbb{K}} \tilde{H}_{j-1}(\Delta|_W).$$

Example 3.5. Consider the simplicial complex Δ pictured below; it consists of a filled in triangle with an edge glued to one of the triangle’s vertices.



It's Stanley-Reisner ideal is $I_\Delta = \langle x_1x_3, x_1x_5, x_2x_5, x_1x_2x_4, x_2x_3x_4 \rangle$ with Betti table

	0	1	2	3
total	1	5	6	2
0	1	.	.	.
1	.	3	2	.
2	.	2	4	2

Using Hochster's formula, we can compute the Betti numbers!

$$\begin{aligned}
b_{1,1}(S/I_\Delta) &= \beta_{1,1+1}(S/I_\Delta) = \sum_{\substack{W \subseteq V \\ |W|=1+1=2}} \dim_{\mathbb{K}} \tilde{H}_0(\Delta|_W) = \underbrace{1+1+1}_{\text{coming from } x_1x_5, x_1x_3, x_2x_5} = 3 \\
b_{1,2}(S/I_\Delta) &= \beta_{1,1+2}(S/I_\Delta) = \sum_{\substack{W \subseteq V \\ |W|=1+2=3}} \dim_{\mathbb{K}} \tilde{H}_1(\Delta|_W) = \underbrace{1+1}_{\text{coming from } x_1x_2x_4, x_2x_3x_4} = 2 \\
b_{2,1}(S/I_\Delta) &= \beta_{1,2+1}(S/I_\Delta) = \sum_{\substack{W \subseteq V \\ |W|=2+1=3}} \dim_{\mathbb{K}} \tilde{H}_0(\Delta|_W) = \underbrace{1+1}_{\text{coming from } x_1x_3x_5, x_1x_2x_5} = 2 \\
b_{2,2}(S/I_\Delta) &= \beta_{1,2+2}(S/I_\Delta) = \sum_{\substack{W \subseteq V \\ |W|=2+2=4}} \dim_{\mathbb{K}} \tilde{H}_1(\Delta|_W) = \underbrace{2+1+1}_{\text{coming from } x_1x_2x_3x_4, x_1x_2x_4x_5, x_2x_3x_4x_5} = 4 \\
b_{3,2}(S/I_\Delta) &= \beta_{1,3+2}(S/I_\Delta) = \sum_{\substack{W \subseteq V \\ |W|=3+2=5}} \dim_{\mathbb{K}} \tilde{H}_1(\Delta|_W) = \underbrace{2}_{\text{coming from } x_1x_2x_3x_4x_5=V} = 2
\end{aligned}$$

Here are some other layouts which might help with understanding how Hochster's formula groups multidegrees into standard graded Betti numbers:

Betti Table Entry	Betti Number	$\tilde{H}_j$	$\Delta _W$	Contributions
$b_{1,1}$	$\beta_{1,2}$	$\tilde{H}_0$	x_1x_5 x_1x_3 x_2x_5	$1 + 1 + 1$
$b_{1,2}$	$\beta_{1,3}$	$\tilde{H}_1$	$x_1x_2x_4$ $x_2x_3x_4$	$1 + 1$
$b_{2,1}$	$\beta_{2,3}$	$\tilde{H}_0$	$x_1x_3x_5$ $x_1x_2x_5$	$1 + 1$
$b_{2,2}$	$\beta_{2,4}$	$\tilde{H}_1$	$x_1x_2x_3x_4$ $x_1x_2x_4x_5$ $x_2x_3x_4x_5$	$2 + 1 + 1$
$b_{3,2}$	$\beta_{3,5}$	$\tilde{H}_1$	$x_1x_2x_3x_4x_5 = V$	$1 + 1$

		0	1	2	3			0	1	2	3
	total	1	5	6	2		total	1	5	6	2
Betti(S/I_Δ) =	0	1	.	.	.	=	0	1	.	.	.
	1	.	3	2	.		1	.	x_1x_5 x_1x_3 x_2x_5	$x_1x_3x_5$ $x_1x_2x_5$	.
	2	.	2	4	2		2	.	$x_1x_2x_4$ $x_2x_3x_4$	$x_1x_2x_3x_4$ $x_1x_2x_4x_5$ $x_2x_3x_4x_5$	$x_1x_2x_3x_4x_5$

4. SOME MORE EXAMPLES

Exercise 4.1. Compute the Betti numbers of S/I_Δ when Δ is the hollow octahedron from the `preliminaries.pdf` notes.

Exercise 4.2. The *star graph* S_n is the complete bipartite graph $K_{1,n}$: it consists of $n - 1$ edges attached at a common vertex (so there are n vertices total).

- Compute the Betti numbers for Stanley-Reisner ring of the star graph S_k for various values of k ; they should have a *very* nice formula. Can you prove expected formula holds in general using Hochster's formula?
- Use `Macaulay2` to compute the Betti numbers of $S/(I_{S_n} + P)$ for different "power ideals" $P = (x_1^{p_1}, \dots, x_{n+1}^{p_{n+1}})$. The code in [helpers.m2](#) might be helpful; in particular, `sunflowerHypergraphIdeal(n,2)` might be helpful.
- Try to prove a similar result for the Betti numbers of `sunflowerHypergraphIdeal(n,k)`.

4.1. Why are we doing this? For showing if certain simplicial complexes have these Lefschetz properties, our primary strategy has been in trying to determine if their Betti tables have (maximal) Koszul tails. This is a condition on the Betti numbers! So, we can try to use Hochster's formula to reverse engineer simplicial complexes with prescribed Betti numbers (in our case: Koszul tails). Putting this all together, for S/I_Δ to have an (n, d) -Koszul tail, we have a system of n equations

(for convenience, $\binom{V}{k}$ is the set of all subsets of V of size k):

$$\left\{ \begin{array}{ll} b_{0,0} = \beta_{0,0} & = 1 \\ \\ b_{1,d} = \beta_{1,1+d} & = \sum_{W \in \binom{V}{1+d}} \dim_{\mathbb{K}} \tilde{H}_{d-1}(\Delta|_W) = n \\ \\ b_{2,d} = \beta_{2,2+d} & = \sum_{W \in \binom{V}{2+d}} \dim_{\mathbb{K}} \tilde{H}_{d-1}(\Delta|_W) = \binom{n}{2} \\ \\ \vdots & \vdots \\ \\ b_{n-2,d} = \beta_{n-2,n+d-2} & = \sum_{W \in \binom{V}{n+d-2}} \dim_{\mathbb{K}} \tilde{H}_{d-1}(\Delta|_W) = \binom{n}{n-2} \\ \\ b_{n-1,d} = \beta_{n-1,n+d-1} & = \sum_{W \in \binom{V}{n+d-1}} \dim_{\mathbb{K}} \tilde{H}_{d-1}(\Delta|_W) = n \\ \\ b_{n,d} = \beta_{n,n+d} & = \sum_{W \in \binom{V}{n+d}} \dim_{\mathbb{K}} \tilde{H}_{d-1}(\Delta|_W) = 1 \end{array} \right.$$

Open Problem 4.3. *Can we solve this system of equations to find examples Δ where the Betti table of S/I_{Δ} has a Koszul tail? How does this change if we want the Koszul tail to be maximal?*